EXPERIMENT - 10

TO DETECT ACID RADICAL PRESENT IN GIVEN SALT SAMPLE BY DRY AND WET WAYS. [SAMPLE A2 & A3]

APPARATUS REQUIRED

a) Test tubes
b) Test tube holder
c) Test tube stand
d) Test tube brushes
e) Wash bottles
f) Bunsen Burner

CHEMICAL REQUIRED

i. Salt sample
ii. dil. HCl
iii. NH_4OH
iv. NaOH
v. Conc. H_2SO_4

THEORY

Qualitative analysis
- It is the branch of chemistry that deals with the study of measurement of quantities of particular constituents present in substance.

Quantitave analysis
- It is the branch of chemistry that deals in the study of identification of constituent. eg:- elements of functional group, radical etc present in a substance.

Dry test
- The test which are performed directly with a dry sample and solid state is called dry tests. Dry tests provide preliminary idea only.

Wet Test

- The test which are performed with the given sample in the form of soln with some suitable solvent are called wet tests. Wet tests provide the confirmation of the result.

Radical

- Radical is an atom or group of atoms having positive or negative charges behave as a single unit during a chemical reaction.

Acid radical

- The electronegative radicals except OH^- are called acid radicals. Eg. Cl^-, SO_4^{--}, NO_3^-, CO_3^{--}, CH_3COO etc.

Salt

- It is the chemical compound formed when the hydrogen ion (H^+) of an acid is replaced by a metal ion or another positive ions. A radical is an atom, molecule or ion that has an unpaired electron in its outer shell, making it highly reactive.

Basic radical

- The electropositive radicals is called basic radical. e.g - Na^+, K^+, Ca^{++}, Al^{3+}.

SAMPLE NO: A_2

State: Solid

Colour: Colourless

Odour: Odourless

Solubility: Soluble in water

OBSERVATION TABLE (Dry test for acid radical)

S.N.	Experiments	Observations	Inferences
1	dil. mineral acid test A pinch of given salt was taken in a clean test tube with Few drops of dil. HCl.	Gas was not evolved	Absence of NO_2^-, S^{--}, SO_3^{--}, and CO_3^{--}.
2	Conc. sulphuric acid test A pinch of given salt was taken in a clean test tube and few drops of conc. H_2SO_4 was added. & warmed.	Gas was not evolved	Absence of Cl^-, Br^- and I^-
3	Dry Nitrate Test In a clean test tube added a little salt & few drops of conc. H_2SO_4 was added	No gas was evolved	Absence of NO_3^-

Note:

If the sample does not respond to any of the above tests, then SO_4^{--} suspected and it's confirmed by performing wet test for SO_4^{--}. In lab generally dry test is not performed

WET TEST FOR ACID RADICAL

A pinch of given salt was taken in a clean test tube & was dissolved in minimum quantity of suitable solvent & diluted by adding more solvents. Then, the following tests were performed.

OBSERVATION TABLE (Wet tests for acid radical)

S.N.	Experiments	Observations	Inferences
1.	1 ml of O.S was taken in a clean test tube. Few drops of $BaCl_2$ soln was added to it.	White ppt was formed.	Presence of SO_4^{--}, CO_3^{--}, SO_3^{--}
2.	The ppt formed from (1) was taken, and dil. HCl was added in excess	The ppt was not dissolved	Presence of SO_4^{--}.

TEST REACTIONS

$SO_4^{--} + BaCl_2 \longrightarrow BaSO_4 \downarrow + 2Cl^-$
White ppt.

$BaSO_4 + HCl \longrightarrow$ No reaction.

SAMPLE NO: A₃

State: Solid

Colour: Colourless

Odour: Odourless

Solubility: Soluble in water

OBSERVATION TABLE (Dry test for acid radical)

S.N.	Experiments	Observations	Inferences
1.	Dil. mineral acid test A pinch of given salt was taken in a clean test tube and few drops of dil. HCl was added.	Gas was not evolved.	Absence of NO_2^-, S^{--}, SO_3^{--} and CO_3^{--}
2.	Conc. Sulphuric acid test	A white dense gas with pungent smell evolved	Analytical presence of Cl^-
	A pinch of given salt was taken in a clean test tube and few drops of conc. H_2SO_4 was added.	A glass rod was taken and it's one end was dipped in $AgNO_3$ soln. This end was brought to the mouth of the test tube White ppt appeared.	Presence of Cl^-

WET TEST FOR ACID RADICAL

A pinch of given salt sample was taken in a clean test tube and was dissolved in minimum quantity of suitable solvent and diluted it by adding more solvent. This was original soln of salt and then following tests were prepared.

OBSERVATION TABLE (Wet test for acid radical)

S.N.	Experiments	Observations	Inferences
1.	1 ml of O.S was taken in a clean test tube. Few drops of dil. HNO_3 was added and boiled for sometime. It was cooled and few drops of $AgNO_3$ soln was added.	There was appearance of curdy white ppt which was soluble in NH_4OH and ppt reappeared on the addition of dil. HNO_3	Presence of Cl^-.

TEST REACTIONS

$AgNO_3 + NaCl \longrightarrow AgCl\downarrow + NaNO_3$
white ppt.

$AgCl + 2NH_4OH \longrightarrow [Ag(NH_3)_2]Cl + 2H_2O$
Soluble (diammine silver chloride)

$[Ag(NH_3)_2]Cl + 2HNO_3 \longrightarrow AgCl\downarrow + 2NH_4NO_3$
white ppt.

RESULT

In this way SO_4^{--} & Cl^- was detected in the given salt sample A_2 & A_3 respectively.

CONCLUSION

Hence, from the above experiment, it can be concluded that acid radical in the given salt sample can be detected both by dry and wet ways.

PRECAUTIONS

1) Personal protective equipment should be worn.
2) Apparatus should be handled with great care.
3) Reagents & chemicals should be used in minimum quantity as far as possible.

VIVA-VOCE

1. Which chemical reagent in used to identify following acid radicals
   i) Cl^- ii) SO_4^{--} iii) NO_3^- ions ?
2. What is formula of brown ring? Give reaction involve in ring test for nitrate.
3. Why borax bead test is not applicable to white salts?
4. How can you identify SO_2 and CO_2 gases liberated from the salt sample when treated with dilute HCl?

1)
i) $Cl^- \rightarrow AgNO_3$ solution.
ii) $SO_4^{--} \rightarrow BaCl_2$ solution followed by HCl
iii) $NO_3^- \rightarrow$ conc. H_2SO_4 and $FeSO_4$

2) The formula at brown ring is $FeSO_4.NO$
Reaction:
$$NO_3^- + H_2SO_4 \rightarrow H_2SO_4^- + HNO_3$$
$$FeSO_4 + H_2SO_4 + HNO_3 \rightarrow 3Fe(SO_4)_3 + 4H_2O + 2NO\uparrow$$
$$FeSO_4 + NO \rightarrow \underset{\text{(Brown ring)}}{FeSO_4.NO}$$

3) White salts donot form coloured metaborates.

4) By performing acidified $K_2Cr_2O_7$ test. SO_2 reacts with acidified $K_2Cr_2O_7$ while CO_2 does not.